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D. A. Baker; J. E. Hammel; F. L. Ribe

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Rotating Plasma Experiments. I. Hydromagnetic Properties

D. A. BAKER, J. E. HAMMEL, AND F. L. RIBE
Los Alamos Scientific Laboratory, University of California, Los Alamos, New Mexico
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Experiments on Ixion III, the Los Alamos rotating plasma device, using electrostatic and magnetic probe techniques are discussed. Analysis of the data indicates the presence of a two-liter annular volume of rotating plasma whose mean mass density is $\sim 1 \times 10^{-8}$ g/cm^3 and whose rotation decays with the supply voltage in ~ 300 μsec. The plasma energy content of ~ 130 joules appears to be nearly equipartitioned between ordered rotational drift motion and random thermal motion. Under typical operating conditions the data show the existence of drift velocities of $\sim 8 \times 10^6$ cm/sec (~ 65 ev for deuterium) and temperatures of the order of 30 ev. The operation of Ixion as a hydromagnetic capacitor in an oscillatory circuit is described.

I. INTRODUCTION

It is well known that the motion of a nonrelativistic charged particle in the presence of uniform, time-independent, mutually orthogonal electric and magnetic fields can be described as a superposition of three simple motions: (1) a cyclotron (Larmor) motion with a constant angular velocity about an axis parallel to the magnetic field, (2) a uniform rectilinear velocity in the direction of the magnetic field, and (3) a uniform rectilinear drift velocity $\mathbf{v}_E$ given by[1]

$$\mathbf{v}_E = \mathbf{E} \times \mathbf{B}/B^2. \quad (1)$$

Of the three velocities, only the latter is completely determined by the fields; the magnitudes of the other velocities depend on the motion of the particle at the time the fields are established. The orbit of the particle is a trochoid and extends to infinity.

If the uniformity condition on the fields is relaxed, the particle orbit can be limited to a finite region of space. The motion in directions normal to the magnetic field can be bounded by establishing the orthogonal electric field radially between two concentric electrodes as shown in Fig. 1. Analysis of the particle orbits in this field configuration shows that to first order the motion of a particle can be decomposed into three motions analogous to those of the uniform field case.[2] The direction of the field-controlled drift is still given correctly by Eq. (1) and is an azimuthal motion about the axis of symmetry. The magnitude of this velocity, however, is given by Eq. (1) only as a first approximation. Sample orbits of a positive ion having no velocity component parallel to the magnetic field are shown in Fig. 1.

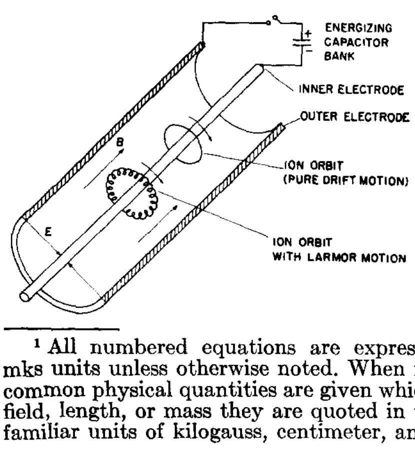


FIG. 1. Simplified schematic diagram showing the Ixion field configuration and sample ion orbits.

In order to confine the motion in the direction of the magnetic field, the field strength may be increased near each end of the cylindrical electrodes, thus placing the particle between two magnetic mirrors. Particle confinement by a magnetic bottle consisting of two magnetic mirrors has been studied extensively by Post and collaborators.[3,4] The effect of adding an orthogonal electric field to produce particle rotation around the symmetry axis has been shown theoretically further to enhance the axial

[1] All numbered equations are expressed in rationalized mks units unless otherwise noted. When numerical values of common physical quantities are given which involve magnetic field, length, or mass they are quoted in the somewhat more familiar units of kilogauss, centimeter, and gram.

[2] The velocity of a nonrelativistic particle with charge q and mass m located in an azimuthally symmetric radial electric field $\mathbf{E}$ and a uniform orthogonal magnetic field $\mathbf{B}$ may be decomposed into three velocities $\mathbf{v} = \mathbf{v}_D + \mathbf{v}_L + \mathbf{v}_z$ (where $\mathbf{v}_D$ is the drift velocity of the guiding center located at a radius r and is given by $\mathbf{v}_D = \mathbf{E} \times \mathbf{B}/B^2$; $\mathbf{v}_L$ is a circular rotation with an angular frequency $\omega = qB/m$ at a radius r_L about the guiding center, and $\mathbf{v}_z$ is a constant drift velocity in the direction of the magnetic field) to a degree of accuracy dependent upon the smallness of the quantities $v_D/\omega r$ and r_L/r compared to unity. In instances where charge densities are present and collisions are negligible, the additional restriction must be imposed that the fractional change of the radial component of the electric field over the distance of the Larmor radius r_L is small, i.e., the particle must not pass through a thin voltage sheath. Under typical Ixion operating conditions, $v_D \approx 8 \times 10^6$ cm/sec, $\omega \approx 4 \times 10^7$ sec^{-1} and $r_L \approx 0.1$ cm for deuterium. The rotating plasma is located at $r > 4.5$ cm so that $v_D/\omega r \lesssim 0.04$ and $r_L/r \lesssim 0.02$.

[3] R. F. Post, Bull. Am. Phys. Soc. **3**, 196(T) (1958).

[4] R. F. Post, in *Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy* (United Nations, Geneva, 1958), Vol. 32, p. 245.

ion containment normally provided by the mirrors.[5,6] The Los Alamos rotating plasma device, Ixion III, being discussed in this paper, is essentially such an electrified magnetic bottle.

The discussion so far has concerned the motion of a single particle. Ixion, on the other hand, produces a rotating plasma having a density of about 10^{15} ion/cm^3. The behavior of the plasma will therefore be influenced by collective effects such as plasma currents, space charges, and particle collisions. Collisions will have the effect (among others) of randomizing the Larmor motion gained by the particles when ionized in the electric field, thus heating the plasma.

This paper and the one immediately following (which will be referred to as paper II) present experimental results obtained since those reported at the 1958 Geneva Conference.[7] In the present paper we shall be concerned primarily with the following: (1) the general operating characteristics of Ixion, including the ionization of injected neutral gas in the crossed fields; (2) the hydromagnetic behavior of the machine as demonstrated by magnetic probes which measure the azimuthal currents arising from centrifugal body forces and pressure gradients; (3) the radial distribution of the electrostatic potential as determined by electrostatic probes; and (4) the characteristics of the machine when operated as an oscillating hydromagnetic capacitor.

In paper II we shall be concerned with the loss mechanisms that give rise to radial, dissipative currents which limit the rotational motion, and hence the back emf, of which the device is capable.

II. OPERATION OF THE IXION DISCHARGE

A. Apparatus

In practice it was found desirable to replace the solid center electrode of the idealized machine shown in Fig. 1 by an arrangement which provides what is essentially a plasma center electrode. Figure 2 is a cross section of the apparatus showing the essential electrical connections as well as the location and construction of various probes to be discussed later. A grounded stainless-steel vacuum chamber 24 cm in diameter, 85 cm long, and with a wall thickness of 2 mm forms the outer electrode (anode)

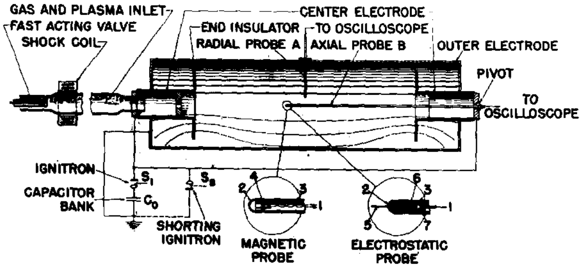


FIG. 2. Cross section of the Ixion device showing construction and probe locations. Probe details are shown in the circular inserts. Key: (1) Output signal leads; (2) Pyrex envelope; (3) stainless-steel electrostatic shield; (4) pickup coil; (5) tungsten pickup electrode; (6) capacitively coupled cup; and (7) insulating spacer.

of the system. Inside the vacuum chamber at each end and separated by 61 cm are two hat-shaped Pyrex insulators whose 19-cm diameter brims are perpendicular to the magnetic lines. Three magnetic lines are indicated in Fig. 2 in the lower half of the vacuum chamber. Inside the cylindrical "crown" sections of the hats are two end electrodes (diameter 6.4 cm). The purpose of the insulators is to force Ixion to draw current through the plasma and only across the magnetic field. If the insulators were not present, the conducting ends of the vacuum chamber would effectively short out the radial electric field. A negative potential is applied through ignitron switches (indicated in Fig. 2 as a single ignitron S_1) to each end electrode by four parallel-connected RG-14/U coaxial cables leading from a 120-μf capacitor bank C_0 capable of being charged to potentials as great as 20 kv. Since electrons can flow readily along the magnetic field lines connecting the end electrodes when plasma is present, a central column along the axis of Ixion is at a uniform negative potential equal to that of the electrodes. This plasma "center electrode" is bounded by a flux surface whose diameter at the ends is that of the electrodes and which is larger at the center by a factor of $\sim$1.5 (the square root of the magnetic mirror ratio).

The vacuum chamber is normally evacuated to a pressure of 2×10^{-6} mm Hg by means of an anti-creep, freon-trapped, oil diffusion pump. Deuterium gas is admitted from the left-hand end (Fig. 2), by means of a fast-acting valve.[8] In the measurements to be reported the valve was operated in such a manner as to fill the system to a pressure from 2 to 5×10^{-3} mm Hg when the pump was valved off. In order to provide initial ionization to initiate

[5] C. L. Longmire, D. E. Nagle, and F. L. Ribe, Phys. Rev. **114**, 1187 (1959).

[6] B. Bonnevier and B. Lehnert, Arkiv Fysik **16**, 231 (1959).

[7] K. Boyer, J. E. Hammel, C. L. Longmire, D. Nagle, F. L. Ribe, and W. B. Riesenfeld, *Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy* (United Nations, Geneva, 1958), Vol. 31, p. 319.

[8] John Marshall, *Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy* (United Nations, Geneva, 1958), Vol. 31, p. 342.

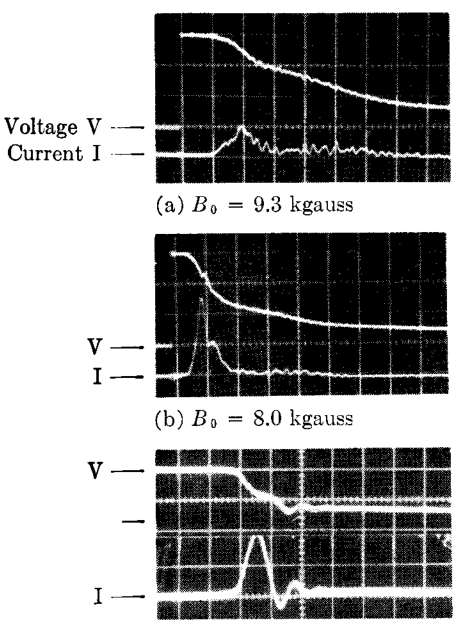
the discharge, a few percent of the injected gas is ionized as it passes through a two-turn magnetic shock coil (Fig. 2) surrounding the Pyrex inlet tube. The plasma so formed is observed to flow at a speed of about 5×10^6 cm/sec along the axial magnetic guide field of the solenoid surrounding the inlet tube and into the vacuum chamber. By varying the time of excitation of the shock coil with respect to the opening of the valve, the plasma can be made to arrive at the center of the machine at various times with respect to the arrival of the slower ($\sim 10^5$ cm/sec) neutral gas.

The magnetic-mirror field is supplied by means of a copper coil cast in epoxy resin surrounding the vacuum chamber. The coil is excited from a 3-kv, 173-kj capacitor bank. This gives a sinusoidal magnetic field with a period of 32 msec and a magnitude as great as 9.5 kgauss at the center of Ixion. The ratio of the magnetic field strength in the mirrors to that at the center is 2.2. Since the main discharge lasts about 300 μsec and is initiated near the time of maximum coil current, the magnetic field is essentially constant during the discharge.

B. General Operating Characteristics

1. Operating Sequence

A typical operating sequence for establishing the Ixion discharge is as follows: The rise of the magnetic field is first initiated, and after a delay of 7 msec the gas valve is opened by a blow from a hammer driven by a magnetic coil. After a further delay of 500 μsec the shock coil is excited, followed 150 μsec later by the application of the capacitor bank voltage to the Ixion electrodes. From this time on the discharge, under normal operating conditions, is self-controlled. The operation of the discharge will be discussed in two phases, (1) the ionization phase and (2) the rotating phase. The first phase is a waiting period ($\sim$100 μsec) after the application of the radial electric field before any measurable current is drawn, and the latter phase begins at the time of current onset.



(a) B_0 = 9.3 kgauss

(b) B_0 = 8.0 kgauss

(c) B_0 = 4.8 kgauss

FIG. 3. Oscilloscope traces of Ixion voltage and current for three values of magnetic field B_0. (a) and (b): V, 2.5 kv/div; I, 10 kamp/div; Sweep speed 50 μsec/div: (c) V, 4 kv/div; I, 20 kamp/div; Sweep speed 20 μsec/div.

2. Ionization Phase

The mechanism by which the main body of gas becomes ionized in the crossed-field configuration is not completely understood. There is evidence that the ionization is accomplished by trapped electrons as in a Penning discharge. The vacuum fields in Ixion are such that with the end electrodes negative, as far as longitudinal motions are concerned, there is an electron potential well at the midplane of the device. Since the magnetic field restricts the transverse electron motion, one would expect a build-up of trapped electrons oscillating about the midplane along field lines. In this Penning type of discharge a few oscillating electrons can efficiently produce further ionizations. The ionization energy can be supplied by the electric field since the vacuum electric and magnetic fields are not totally orthogonal. The Penning discharge hypothesis is supported by the following experimental observations: (1) The ionization of the main body of gas does not occur without an applied magnetic field of at least a few hundred gauss. (2) When Ixion is operated with reversed polarity, the ionization does not take place within the time requirements of the pulsed magnetic field. The ionization, if it occurs, is delayed many milliseconds. A further effect (not necessarily confirming the above hypothesis) noted was that the delay time between the bank switching and the onset of Ixion current is increased as the amount of injected gas is reduced.

3. Rotational Phase

Typical Ixion voltage and current oscillograms are shown in Fig. 3. Traces are shown for three different values of applied magnetic field (B_0 is the value of the vacuum field at the midplane). The voltage V_0 = 7.5 kv is switched on, remains at full value during the ionization phase, and then drops as a current pulse is drawn, setting the plasma into rotation. During this process currents flow radially from the anode walls and axially to the end electrodes, giving rise to azimuthal $\mathbf{J} \times \mathbf{B}$ forces whose associated torque accelerates the plasma into rotation. The ionization processes begun earlier are continued as the initial pulse of current is drawn. The voltage drops to a value equal to the back emf of the rotating plasma which depends on the magnetic field strength as demonstrated by Figs. 3(a) and 3(b). After the initial pulse (10 to 40 kamp) the current drops to $\sim$2 kamp. Thereafter the rotation is sustained for $\sim$300 μsec, and the capacitor bank is drained as it supplies energy losses. It is found that the discharge decays in such a fashion that it leaves $\sim$1 kv on the energizing capacitors after the plasma is gone. The operation just described will be referred to as the *voltage-holding* mode. If the magnetic field is decreased sufficiently the back emf is not sustained; the current and voltage oscillate as shown in Fig. 3(c). Such an *oscillatory* mode is also produced if voltages in excess of $\sim$7.5 kv are applied to Ixion.

The plasma rotation was demonstrated spectroscopically, as reported at the Geneva Conference,[7] by observing the Doppler shift of a spectral line from rotating carbon impurity ions. Typical data at $V_0 = 7$ kv show rotational velocities of 4×10^6 cm/sec at a radius of $\sim$11 cm.

There is an uncertainty concerning the composition of the rotating plasma. The vacuum system used is not of particularly high integrity and contains deuterium and impurity molecules loosely absorbed on its walls, easily equal to the number of deuterium molecules injected in a given discharge. These experiments were intended to operate at plasma number densities in the region of 10^{15} cm^{-3} or less. Under these conditions the discharge may absorb sufficient wall gas to affect the ion density, which must therefore be measured in the plasma rather than deduced from the quantity of injected gas. Spectrograms of the discharge show strong deuterium lines and appreciable additional lines of silicon, carbon, and oxygen typical of an oil pumped system.

III. ELECTROSTATIC-POTENTIAL MEASUREMENTS

A. Technique

Electrostatic-potential measurements as a function of radius were made with a capacitively coupled high-impedance electrostatic probe. The basic construction of the probe is that shown in Fig. 2. The voltage signal is picked up by direct plasma contact with a tungsten wire protruding 3 mm beyond a glass seal at the end of a 7-mm diameter Pyrex tube. The pickup electrode is then capacitively coupled to an oil-filled cup using insulating spacers. This coupling capacity proves to be 4.5 $\mu\mu$f. The cup side of this capacitor is connected to ground through a 0.1 μf high-quality capacitor. The voltage across the latter capacitor is measured with an oscilloscope, so that the arrangement is in essence a capacitive voltage divider. The probe is inserted through a sliding vacuum seal and is manipulated in a fashion similar to that described in the next section on magnetic probes. Two side ports, one at the midplane and the other 19 cm from the midplane, as well as the pivoted end port are used to obtain radial voltage distributions (cf. Fig. 2). All measurements reported in this section were made with an applied magnetic field B_0 of 9.3 kgauss.

B. Voltage Distributions during the Ionization Phase

We first consider measurements of the voltages present during the early ionization phase before the onset of the Ixion current pulse. During this period the probe signals exhibit large amounts of "hash". When the probe tip is positioned about 1 cm from the outer wall electrode, the signals indicate a probe voltage which oscillates between small positive values and the full applied voltage of -7.5 kv. These signals are not periodic but change in times of the order of 0.05 μsec.

C. Voltage Distributions during Rotation

The radial voltage distribution, 100 μsec after the onset of the charging current, as deduced from the electrostatic probe signals, is given in Fig. 4. At this time the Ixion voltage has dropped from its initial value of 7.5 kv to 4.75 kv. The data are shown for the midplane and for a plane 19 cm from the midplane. It was possible to obtain data from the pivoted axial probe from the center out to a radius of $\sim$4 cm without observably affecting the Ixion operation. Likewise, the side probe produced good data up to 2.5 and 5.5 cm from the wall at the midplane and the 19-cm plane, respectively. In Fig. 4 the solid points represent data obtained at probe positions for which the Ixion current and voltage characteristics are not observably changed. The remainder of the points were obtained under conditions for which the back emf is appreciably reduced by the effect of the probe and the Ixion operation affected. The latter points are not necessarily to be interpreted as having any relation to

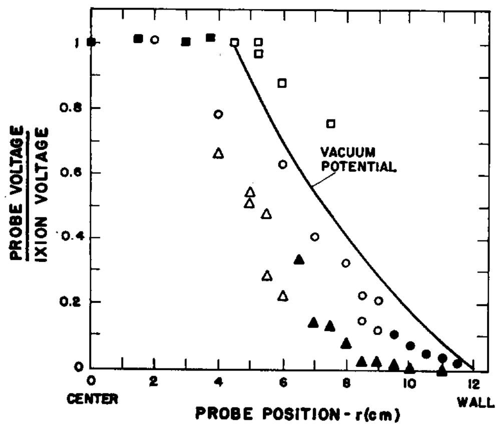


FIG. 4. Radial voltage distributions as obtained with an electrostatic probe. The circular and square data points give the voltage at the midplane obtained with the radial probe and pivoted axial probe, respectively. The triangular points give the voltage data at a station 19 cm from the midplane. Open points represent data taken at positions where serious probe perturbations were evident. The experimental error is approximately the size of the data symbols.

the voltage distribution under normal unperturbed rotation. They are given to show that the probe does indeed see a larger fraction of the voltage change as it is inserted deep into the active regions of the discharge. For comparison, the vacuum logarithmic potential curve is shown which assumes an effective center rod of 4.5-cm radius, as predicted from the measured vacuum magnetic field, assuming the flux surfaces to be equipotentials.

D. Discussion

A few remarks on electrostatic probes are in order. The common objections which are raised to the use of electrostatic probes in plasma devices are as follows: (1) Temperature effects can cause an electrode to acquire a net charge due to the different emission and mobility characteristics of ions and electrons. (2) There may be voltages induced by changing magnetic fields giving rise to possible misinterpretations of the data. (3) The probe may alter the discharge in such a way that the fields sensed by the probe materially differ from those normally present without the probe perturbation. Consideration of the foregoing statements, as applied to the present experiment leads to the following conclusions: (1) Temperature effects can only give rise to charge separation potentials of the order of kT/e, which in the present case is on the order of a few tens of volts. We are primarily concerned with measurements of potentials between hundreds of volts and kilovolts, and the percentage error due to temperature effects should therefore be small. (2) The data are taken at a time when the Ixion currents are relatively small, the main portion of the current pulse having subsided. The current is not changing rapidly, and induced voltages should be small compared to the voltage of interest. (3) Uncertainties concerning the effects of probe perturbations on a plasma are never completely dispelled for any type of material probe. However, the following precautions were taken: (a) When the probe produced observable effects on the discharge, the data were either discarded or the effect noted. (b) The small capacitive coupling was used to reduce plasma loading effects. (c) The probes were made small in order to minimize the perturbations.

With the assumption that the measurements give a true evaluation of the voltage distribution over the range covered by the solid points on Fig. 4, one can make the several interesting observations: (1) The electric field near the wall is small. This voltage suppression may be a boundary-layer phenomenon connected with an influx of particles from the wall or direct plasma-wall interactions. The lowered electric field near the wall has the consequence of implying higher electric fields at interior radii in order to account for the full applied voltage. One cannot set an upper limit on the magnitude of the electric-field intensity from the data. From the data one deduces that an average electric field of 740 v/cm must exist in the region between radii of 3.75 and 9.5 cm, where reliable data could not be obtained. This result implies the existence of regions in which the plasma drift velocity is at least 8.0×10^6 cm/sec. From the voltage profile it appears that a major portion of the rapidly rotating plasma lies in an annular region which extends from the effective center rod out to a 9.5-cm radius. The reduced value of the electric field near the wall is in accord with the lower value (4×10^6 cm/sec) of drift velocity which was deduced from the Doppler shift[7] of a spectral C II line measured at a radius of 11 cm. (2) Although one cannot probe adequately in the neighborhood of a 5-cm radius without seriously affecting the discharge, the indications are that the effective center rod is near the 4.5-cm radius one would predict assuming that magnetic flux surfaces and equipotential surfaces coincide. (3) The measured vacuum flux surface that grazes the outside electrode is 2.5 cm from the wall at a position removed 19 cm from the midplane. It is at this radius and axial location that a probe begins to detect a voltage as it is moved in from the wall (see Fig. 4). This result is a confirmation of the hypothesis that flux surfaces are indeed electrostatic equipotentials.

These results show that the electric field penetrates the body of the plasma and that, during rotation, there is no anode sheath of the type discussed by Longmire.[9] On the basis of crossed fields in infinite parallel-plate geometry, in the guiding-center drift approximation, Longmire predicts the formation of an anode sheath across which a major fraction of the applied voltage appears. The thickness of the sheath in centimeters is given by

$$x = 3.3V^{\frac{1}{2}}/B, \qquad (2)$$

where V is the potential between the plates in volts and B is the magnetic field in gauss. Applying this formula to the present case, one predicts a sheath thickness of 0.25 mm as a first approximation for our cylindrical geometry. The only possible evidence of voltage sheath formation at the outer cylindrical electrode appears during the early ionization phase prior to the onset of the current pulse. The rapid appearance and disappearance of the full voltage at a position $\sim$1 cm from the wall has the possible interpretation of representing a voltage sheath which repeatedly forms and disintegrates at roughly a 10 Mc/sec rate. The stability of the Longmire sheath was studied by Ebel and Treiman[10] with the conclusion that it is unstable. Thus, from a theoretical standpoint a quiescent voltage sheath is not be be expected. The presence of a cylindrical oscillating sheath cannot be ascertained definitely by the single-probe experiment described here. Time-correlated measurements with several probes would presumably answer the question.

IV. MAGNETIC-FIELD MEASUREMENT

A. Introduction

The rotating plasma in Ixion may be expected to produce a weakening of the magnetic field, or diamagnetism, inside the device. This arises essentially from the centrifugal force density of the plasma which causes the magnetic flux surfaces to be forced outward from the axis of the machine. For uniform fields the guiding center $\mathbf{E}\times\mathbf{B}$ drift velocities of ions and electrons are equal. With a radial electric field, however, the presence of rotation gives rise to centrifugal forces on the ions and electrons which cause the drift velocities to differ, resulting in a net azimuthal current which gives rise to the diamagnetism. There is in addition a component of the diamagnetism which arises from gradients of the plasma pressure.

In the following are presented the results of magnetic-probe measurements from which determinations were made of the plasma density and spatial extent, as well as the energy contained in ordered rotational and random Larmor (thermal) motion.

B. Experimental Method

Measurements of the magnetic field present during the holding mode were made by means of magnetic probes. The radial and axial distributions of the axial component of magnetic field were measured at the midplane and along the symmetry axis, respectively, by means of two magnetic pickup loops A and B positioned as shown in Fig. 2. Further measurements of radial distributions a few centimeters to either side of the midplane would yield definite knowledge concerning the axial length of the rotating plasma. The latter measurements were not accomplished because of restrictions imposed on the port locations by the magnetic-field coil.

The probes consisted of small, electrostatically shielded coils (effective area $\sim$7 cm^2) encased in 7-mm diameter quartz envelopes. The probe signals were time integrated and displayed on an oscilloscope, thus indicating changes from the applied vacuum magnetic field. The radial probe A, could be inserted through the wall at the midplane inward to any desired radius. It was found, however, that the discharge was drastically altered (an increase of current and a lowering or annihilation of the back emf of Ixion) if the probe was inserted farther than $\sim$3 cm from the wall. The probe B could be located at any desired point on the axis, and in addition was pivoted in such a fashion that it allowed radial motions at the midplane out to radii of 6.7 cm. Data were obtained with the B probe along the axis and radially on the midplane

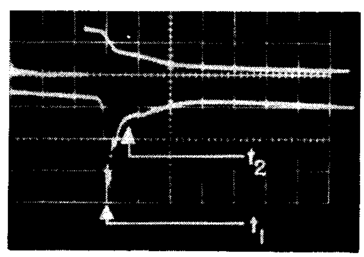


FIG. 5. Top trace: Ixion voltage. Bottom trace: integrated signal ΔB from a magnetic probe located on the symmetry axis. The time of maximum diamagnetism t_1 is also the time of maximum external current. The time t_2 indicates the termination of the large current pulse. Sweep speed, 100 μsec/cm; B_0 = 7.0 kgauss; maximum ΔB = $-$1.9 kgauss; initial voltage V_0 = 7.5 kv.

[9] C. L. Longmire, *Plasma Physics*, Chap. III (to be published).

[10] M. E. Ebel and S. B. Treiman, Los Alamos Scientific Laboratory Rept. LA-2408 (1959), p. 16.

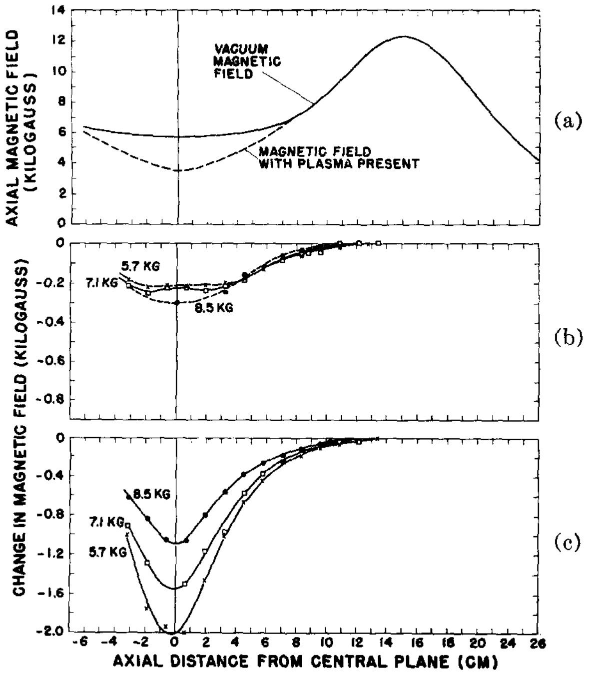


FIG. 6. Axial distributions of magnetic field. The deformation ΔB of the vacuum magnetic field at the time of peak diamagnetism is shown in (a). The values of ΔB at the early and late times t_1 and t_2 are shown in (c) and (b), respectively.

out to distances of $\sim$5 cm from the axis without noticeably changing the nature of the discharge.

C. Axial Distribution of Magnetic Field

A typical signal from probe B located on the axis together with an Ixion voltage trace is shown in Fig. 5. There is a large diamagnetic signal at the time of the voltage change t_1 and then a tail which decays with the voltage. The curves presented in the following figures were obtained from measurements of the probe signals at the two times t_1 and t_2 designated in Fig. 5. A general picture of the axial distribution of the plasma diamagnetic effect is shown in Fig. 6(a). The solid curve shows the axial component B_z of the applied vacuum field as measured along the axis of symmetry. The dashed line is the corresponding curve as deduced from magnetic probe signals at the time of peak diamagnetism. More detailed plots of the axial probe data are given in Figs. 6(b) and 6(c). A family of curves with the applied vacuum magnetic field on the midplane B_0 as a parameter, showing the change in the magnetic field ΔB_z as a function of axial distance from the midplane as obtained from probe signals at the time of maximum signal t_1, is given in Fig. 6(c). Figure 6(b) shows the corresponding curves at the later time t_2.

D. Radial Distribution of Magnetic Field

The radial distribution of ΔB_z on the midplane at the time t_1 for three different values of applied field B_0 is shown in Fig. 7(a). The corresponding curves at the later time t_2 are shown in Fig. 7(b). The values of t_1 and t_2 given in Fig. 7 are measured from the time of onset of the current pulse. The circular data points near the wall were obtained with the radial probe A while the points designated by squares were obtained by pivoting the axial probe B radially outward from the axis. The portions of the curves in the vicinity of an 8-cm radius lack data points because probes in this region of the plasma seriously affect the Ixion holding mode of operation, and useful probe data can not be obtained. In contrast to the axial probe signal which gives well-defined traces, the radial probes display extremely hashy signals at large radii, indicative of a nonquiescent plasma boundary. The circular points in the radial plots represent the average fields upon which the noisy disturbances are superposed.

The effect on the magnetic field caused by sudden removal (shorting) of Ixion voltage is displayed in Fig. 8. The short was applied by firing ignitron S_s of Fig. 2. The probe signal on the axis and the voltage across Ixion are shown in Fig. 8(a). The time of shorting is designated by the arrow. Note that on the axis the diamagnetic signal disappears immediately with the voltage. Figure 8(b) shows a

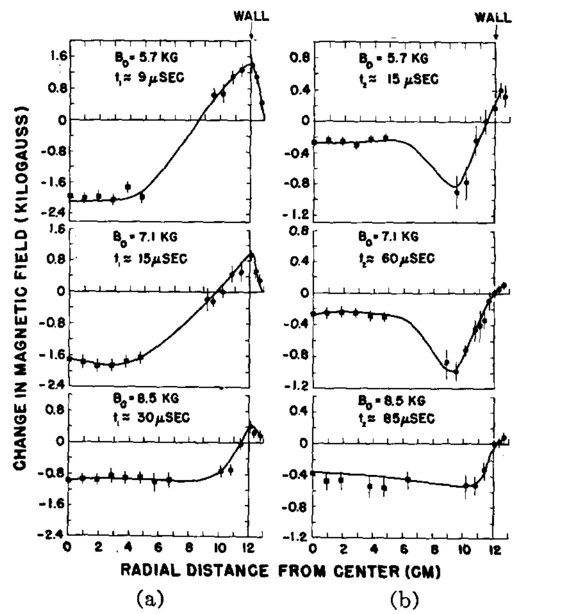


FIG. 7. Radial distributions of the deviation ΔB of the axial component of magnetic field from its vacuum value at the midplane. (a) corresponds to the early time t_1 and (b) to the later time t_2.

simultaneous probe signal on the axis and at a 10-cm radius and Fig. 8(c) shows the same signals when no short is applied. One can see that at the outer radius the signal persists for some 20 μsec after the voltage removal. The change in the radial distribution of B_z at the midplane resulting from suddenly shorting Ixion is given in Fig. 9. The field distributions immediately before and after the time of external shorting are given by the solid and dashed curves, respectively.

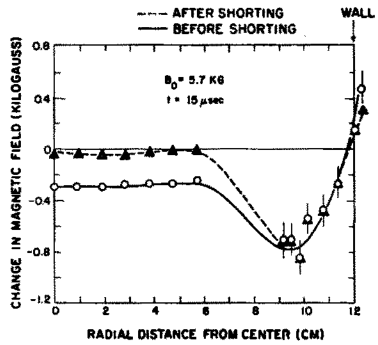


FIG. 9. Radial distribution of the axial component of diamagnetic probe signal ΔB before and after a short is applied to Ixion.

E. Discussion

1. *Axial Field Distributions*

Inspection of the axial data of Fig. 7 leads one to the conclusion that the active portion of the plasma is confined to a region between the mirrors of the magnetic bottle. This confinement can arise from two processes: (1) ordinary magnetic-mirror confinement with an enhancement resulting from plasma rotation[5,6] and (2) the pinch effect of the B_θ fields arising from the axial current flow to the end electrodes. The latter effect is a compression of the plasma radially toward the axis and axially toward the midplane and is most important when large plasma currents are flowing. The exact length of the active plasma region cannot be deduced uniquely from a knowledge of B_z on the midplane and axis. A detailed deduction of the space and time dependence of plasma current densities would require field data throughout the volume. One can, however, gain some insight regarding the possible length of the active region by making simple plasma current-density models and determining what their dimensions must be to conform to the field data. Each axial field plot at the time of current peak [Fig. 7(c)] corresponds very closely to the field on the axis of a thin ring of azimuthal current. For example, the curve in Fig. 7(c) for $B_0 = 5.7$ kgauss reproduces very closely the field of a thin ring current of 14.3 kamp at a radius of 4.5 cm. The other two curves can similarly be represented by ring currents at approximately the same radius but with current values reduced in proportion to their field values. It is interesting to note that this simplified comparison puts the azimuthal currents near the periphery of the predicted effective center rod. One must also include, however, the effects of the induced currents in the outside cylindrical electrode. When these effects are considered, one is still led to the conclusion that the plasma current must be an annulus which cannot be more than a few centimeters thick in the axial direction in order to give the rapid decrease of the axial field shown in Fig. 6(c). In view of this conclusion, reliable pressure balance calculations cannot be made from the B_z data at the time of the current peak. The reason is that pressure terms involving B_r and B_θ and their derivatives become important when the active plasma is highly compressed axially.

The axial distributions of magnetic fields at later times [Fig. 6(b)] are broader and indicate that the current distributions have spread out axially from those during peak current. This is to be expected since the currents at these later times are much lower and the B_θ compression has reduced to a relatively low value.

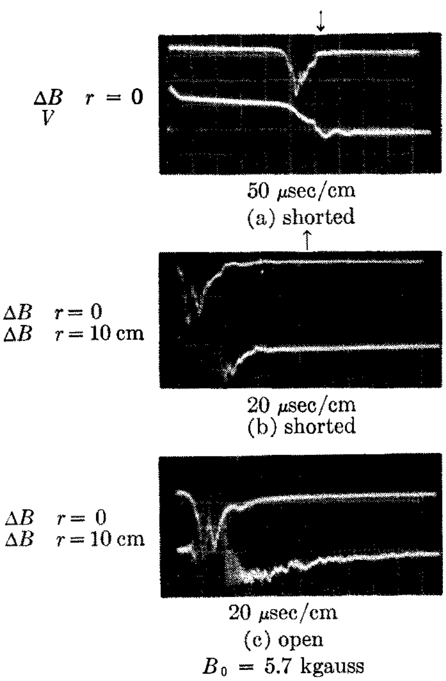


FIG. 8. Oscillograms showing magnetic probe signals ΔB when the Ixion is shorted. The early part of the probe signals at the 10 cm radial position are hashy and do not appear clearly after reproduction. The time of shorting is shown by an arrow in (a) and (b). The corresponding signals when no short is applied are shown in (c).

An estimated effective axial plasma length of $L = 10$ cm at the time t_2 will be used in later calculations. Any number assigned to the length from the axial field distributions will be in some doubt in view of such complicating unknowns as the distribution of the induced currents in the outer cylindrical electrode and the shape of plasma volume.

2. Radial Field Distributions

In drawing curves through the data points obtained at the midplane (Fig. 7), one has to make a guess as to the field behavior in the region where data could not be obtained. If it is assumed that the ΔB does not change sign in this region, it is clear from the data that flux is not conserved inside the outer cylindrical electrode. This effect becomes more pronounced at later times. This apparent loss of flux may be accounted for, within the experimental error, by combination of three effects. (1) The resistivity of the outer stainless steel electrode is high enough to allow considerable flux to escape through it. Observations with a loop around the outside of the cylindrical electrode showed that a sizeable fraction of the missing flux escapes within 50 μsec. (2) Calculations[11] have shown that the field penetrates into the metal of the electrode in a time much shorter than the L/R time of the cylinder. For the present case $L/R \sim 200$ μsec and the time for penetration of field into the metal is ~ 20 μsec. Thus some flux can be lost into the skin of the metal electrode. (3) The hole in the wall through which the probe was inserted has the effect of reducing the field near the hole. For example, a small hole in the surface of a long sheet solenoid will reduce the field at the center of the hole (as approached from inside the cylinder) to $\frac{1}{2}$ the field when no hole is present. This effect is appreciable at positions within a hole diameter from the wall. For the radial probe data one can expect reduced field readings within about a centimeter from the wall. Since cylindrical symmetry is assumed in calculating the flux, this decreased field near the hole would cause one to deduce too high a value for the flux loss.

The nonconservation of flux is expected to have adverse effects on radial confinement. The leakage of magnetic field from the volume undoubtedly allows a portion of plasma to relax to the wall. Lehnert[12] has suggested that this effect may contribute the voltage limiting phenomenon discussed in paper II. It would, therefore, seem advisable to make the outer cylindrical electrode from a good conductor such as copper. In the present experiment, the electrode conductivity was necessarily limited in order to be able to establish the field using a pulsed current source.

The radial plots of magnetic field show the following characteristic property: At early times [Fig. 7(a)] the axial component of magnetic field appears to vary monotonically with radius, while at later times [Fig. 7(b)] the distributions have both positive and negative slopes. It is believed that in the first case the distributions are dominated by the diamagnetic effects associated with plasma rotation.[13] At later times the rotational energy is reduced and the effects of Larmor (thermal) energy make themselves manifest in the shape of the field distribution. This contention is supported by the fact that when Ixion is shorted, the field on the axis disappears but remains at the larger radii (see Figs. 8 and 9). This effect is interpreted as follows: When the short is applied the radial electric field is removed, the rotation stopped, and the diamagnetic signal at the center due to plasma rotation vanishes. The Larmor motion, being random in phase, remains and produces the field that persists at the larger radii. The time decay of the diamagnetic signal at these radii is indicative of such effects as the charge exchange and mirror losses when no electric field is present.

3. Deduction of Plasma Energy Content from Pressure Balance

A precise computation of the plasma energy requires a knowledge of the magnetic field throughout the active volume of the discharge. A rough estimate of the energy density at the midplane can be made from the data for the time t_2 if the following simplifying assumptions are used: (1) Collision rates are such that by the time t_2, at which the large current pulse has subsided, the plasma is adequately described by a scalar pressure p and an average drift velocity $\mathbf{v}_D$. (2) A high degree of axial and radial confinement exists so the $\mathbf{v}_D$ has negligible axial and radial components. (3) The system has azimuthal symmetry. (4) The plasma is in a quasi-steady state in the sense that inertial terms arising from the explicit time dependence of $\mathbf{v}_D$ are negligible. (5) Charge separations are such that electrostatic forces are negligible compared to the magnetic

[11] J. C. Jaeger, Phil. Mag. **29**, 18 (1940). We are indebted to E. J. Stovall for carrying out the numerical computations.

[12] B. Lehnert, "Voltage characteristics of a rotating plasma" (to be published).

[13] The diamagnetic effects arising from centrifugal body forces associated with $\mathbf{E} \times \mathbf{B}$ drifts in cylindrical geometry are discussed theoretically in reference 7.

forces on the plasma. With these assumptions, the only inertial term of consequence is the centrifugal force arising from the azimuthal drift velocity $\mathbf{v}_D$. Equating the sum of the radial forces per unit volume to zero yields the following equilibrium equation:

$$\rho v_D^2/r + [\mathbf{J} \times \mathbf{B} - \nabla p]_r = 0, \qquad (3)$$

where ρ is the plasma mass density and the r subscript denotes the radial components. We shall restrict ourselves to the midplane where from symmetry B_θ and J_z are zero so that the radial component of $\mathbf{J} \times \mathbf{B}$ becomes $J_\theta B_z$. From Maxwell's equation we have $J_\theta = \partial H_r/\partial z - \partial H_z/\partial r$. The derivative of H_r is not obtainable from the data. For the present calculation we shall neglect this term assuming that by the time t_2 the plasma is spread axially in such a fashion that the second term dominates. We can then write the pressure balance equation in the following form:

$$(\partial/\partial r)[(B_z^2/2\mu_0) + p] = \rho v_D^2/r. \qquad (4)$$

One would expect that shorting Ixion would remove its internal electric field and hence the drift velocity ($v_D \approx E/B$) and the right-hand side of the pressure balance equation would be zero. Integrating with respect to radius then gives a pressure distribution equation

$$p(r) = p(R) + (1/2\mu_0)[B_a^2(R) - B_a^2(r)], \qquad (5)$$

where R is the radius of the outer electrode and the a subscripts denote B_z values just after the short. Substituting the field values given by the dotted curve in Fig. 9 and assuming that the pressure at the wall $p(R)$, is nil (nkT at the wall will be small since the wall will cool the adjacent plasma), one obtains a pressure distribution function. This function is large where $|\Delta B|$ is large and therefore gives a hump of pressure lying between radii of 6 and 12 cm and reaching a maximum value of $\sim 4 \times 10^{-2}$ joules/cm^3 at $r = 9.5$ cm. This statement assumes that the dotted curve represents the data in the unprobed region. It is quite possible that the Larmor energy density may be considerably higher than the above number in the region where accurate probe data were not obtained. Assuming azimuthal symmetry, numerical integration of $p(r)$ over the Ixion cross section gives the Larmor linear energy density at the midplane just after the short is applied:

$$W_\perp = \int_0^R p(r) 2\pi r \, dr = 9 \text{ joules/cm}. \qquad (6)$$

It can be shown[14,15] that if the time of changing a uniform electric field normal to a magnetic field is long compared to the Larmor period of the charged particles in the fields, the Larmor energy of the particles is not appreciably altered by the changing field. Let us consider the validity of applying this statement to the present case of shorting Ixion. The Larmor period is ~ 0.2 μsec for deuterium ions and a factor of m/M shorter for electrons. The removal of the voltage by applying the short takes several microseconds; hence to the extent that $\mathbf{E}$ is uniform over distances comparable to Larmor radii, shorting Ixion should leave the Larmor energy unchanged. There is a small effect caused by the electric fields induced by the changing of the plasma diamagnetism as the rotation is stopped. Since the time of the magnetic field change is large compared to a Larmor period, we can estimate this effect using the single-particle adiabatic invariant[16] $\mu = w_\perp/B$. In this expression $w_\perp = \frac{1}{2}mv_\perp^2$ is the particle kinetic energy associated with the velocity normal to the magnetic field. The invariance of the magnetic moment μ requires $\Delta w_\perp/w_\perp = \Delta B/B$. Applying this formula to the present case, one finds that the fractional change in the magnetic field produced by shorting and hence in the Larmor energy is $\sim 4\%$.

Using the fact that the Larmor energy is essentially the same just before and after the time of shorting, one can now substitute the pressure distribution that was obtained from Eq. (5) (using the postshort data) into Eq. (4) to compute the rotational energy density before shorting:

$$w_r = \tfrac{1}{2}\rho v_D^2 = (r/4\mu_0)(\partial/\partial r)[B_b^2(r) - B_a^2(r)], \qquad (7)$$

where $B_b(r)$ and $B_a(r)$ are the magnetic fields present just previous and after the application of the short. Inspection of Fig. 9 shows the two field distributions to be constant for $r < 5$ cm and nearly equal for $r > 9.5$ cm. Equation (7) therefore implies the existence of a rotating annulus of plasma in the region 5 cm $< r <$ 9.5 cm which, reasonably enough, is the region in which probes seriously effect the rotation. This is also the region in which the electrostatic probe measurements indicate a high electric field. If an average derivative for Eq. (7) is approximated from the measured field values at $r = 5$ and 9.5 cm one obtains 10^{-2} joules/cm^3 as an order of magnitude estimate for the rotational energy density in the annular region.

To estimate the rotational energy per unit length

[14] Reference 9, Chap. II.

[15] S. Chandrasekhar, *Plasma Physics* (University of Chicago Press, Chicago, 1960), p. 20.

[16] Reference 9, Chap. III or reference 15, p. 24.

at the midplane, Eq. (4) is multiplied by πr^2 and integrated from 0 to R (the radius of the outer wall). After an integration by parts one obtains

$$W_r + W_\perp + W_f = W_w, \tag{8}$$

where

$$W_r = \int_0^R \tfrac{1}{2}\rho v_D^2 2\pi r\, dr \tag{9}$$

is the plasma rotational drift energy per unit length, and

$$W_f = \int_0^R \frac{B^2}{2\mu_0} 2\pi r\, dr \tag{10}$$

is the corresponding magnetic field energy density. $W_\perp$ is the Larmor energy per unit length defined by the integral in Eq. (6), and

$$W_w = \pi R^2[p(R) + (1/2\mu_0)B^2(R)] \tag{11}$$

is the product of the Ixion cross-sectional area and the sum of the Larmor and field energy densities at the wall. For the present case, using the preshort data of Fig. 9 in Eqs. (10) and (11), one obtains $W_f = 50$ joules/cm, and $W_w = 65$ joules/cm [assuming again that $p(R)$ is negligible]. Using these values and the previously determined value $W_\perp = 9$ joules/cm in Eq. (8) yields $W_r = 6$ joules/cm. This number depends sensitively on the measured field values at the wall and therefore contains errors resulting from the probe port perturbation as mentioned previously. The rotating energy will be obtained from the probe data in a different way in the next section and also from a direct measurement discussed in paper II. Equation (8) allows the computation of the sum of the Larmor and rotational

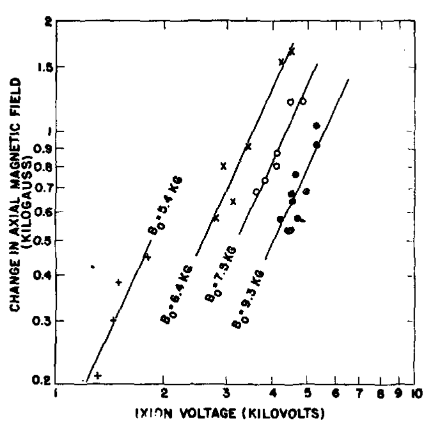


FIG. 10. A plot of plasma diamagnetic probe signal ΔB at the center of Ixion vs the corresponding Ixion voltage for four values of applied magnetic field. Data points with the same symbol represent data taken at different times during the discharge for a given applied magnetic field B_0.

energy densities from the radial field plots of Fig. 7(b) without reference to a shorting experiment. For the case just discussed $B_0 = 5.7$ kgauss, $W_r + W_\perp = 15$ joules/cm. The corresponding values deduced from the other two field plots of Fig. 8(b) are 12 and 13 joules/cm for B_0 values of 7.1 and 8.5 kgauss, respectively. One is therefore led to the interesting conclusion that the charging current pulse energizes the plasma to nearly the same amount independent of the value of the magnetic field over the range of fields studied. This result is reflected in another way by the value which the Ixion voltage V assumes immediately after the current pulse, i.e., at the time t_2. The experimental values of V are 2.45, 3.1, and 3.4 kv for B_0 equal to 5.7, 7.1, and 8.5 kgauss, respectively; the ratio of V/B is nearly constant at $\sim$0.4 v/gauss. The electrostatic probe data suggest that one can approximate E by $V/\Delta r$, where $\Delta r \approx 5$ cm is the radial thickness of the region of rotating plasma as estimated from both the electric and magnetic-probe measurements. One then computes $v_D \approx E/B_0 \approx 8 \times 10^6$ cm/sec for the drift velocity assumed by Ixion just after the charging current pulse is drawn for the range of magnetic fields studied.

For purposes of comparison later, estimates of the total Ixion energies rather than linear energy densities are needed. These can be obtained by multiplying the midplane linear densities above by the effective length of 10 cm estimated earlier.

F. Voltage and Magnetic-Field Dependence of the Plasma Diamagnetism

1. Experimental Data

The dependence of the diamagnetic signal on the axis at the midplane upon the magnetic field and Ixion voltage are displayed in Fig. 10. The probe data giving $-\Delta B$ as a function of Ixion voltage are plotted with the applied magnetic field B_0 as a parameter. The initial bank voltage was held constant at 7.5 kv and the probe data were plotted against Ixion voltage at times corresponding to near the peak of charging current pulse until approximately 60 μsec thereafter.

2. Analysis and Interpretations

From the slopes of these curves one finds that ΔB varies as the square of the Ixion voltage V. The data also show that ΔB varies approximately as $1/B_0^3$. It will now be demonstrated that these dependences are in accord with the premise that the mean density of the rotating plasma remains constant over the range of fields and times represented by the data of Fig. 10. This will be done by determining the electric and magnetic field dependence of the azimuthal currents responsible for the production of the field change on the axis. In the rotational region the currents are primarily azimuthal and radial while **B** is nearly axial. Thus it is valid to neglect $J_z B_\theta$ in comparison with $J_\theta B_z$ when computing the r component of $\mathbf{J} \times \mathbf{B}$. Equation (3) can then be solved for J_θ giving

$$J_\theta = (1/B)(\partial p/\partial r - \rho v_D^2/r). \quad (12)$$

The field change ΔB on the axis is expressable as a volume integral of the Biot-Savart law in which J_θ is a distributed source. The shorting experiment discussed above has shown that the ΔB on the axis results solely from currents related to plasma rotation and not from those associated with plasma pressure. Thus in computing ΔB on the axis from the current density, the first term in the brackets of Eq. (12) will give no contribution to the integral and can be dropped. Since to a good approximation $v_D \approx E/B$, one can compute ΔB on the axis in terms of a volume integral whose integrand contains (as a factor) an azimuthal drift current density given by

$$J_d = -\rho E^2/rB^3. \quad (13)$$

Noting that $B \approx B_0$ and $E \propto V$ in the drift region, one then concludes that $\Delta B \propto V^2/B_0^3$ provided ρ is independent of V and B_0.

A detailed knowledge of J_d is not at hand, but we can estimate the average plasma mass density by assuming the current density to be distributed over an annulus which from probe data is estimated to lie between radii $a = 4.5$ cm (the radius of the effective center rod) and $b = 9.5$ cm and to have an effective length $L = 10$ cm. The electrostatic probe data indicate that a major portion of the Ixion voltage appears between the above radii. We shall approximate E_r by assuming it is constant over the current region and given by $E_r = V/(b - a)$. In the integration we can replace $B(r)$ by B_0 without introducing a serious error since after the current pulse the fractional deviation of B from its vacuum value is small. One may now compute the field ΔB at the center of the annulus with the current density given by

$$J_d = -\rho V^2/[rB_0^3(b - a)^2]. \quad (14)$$

The result is

$$\Delta B = -\mu_0 \bar{\rho} F V^2/B_0^3, \quad (15)$$

where $\bar{\rho}$ is the mean value of mass density in the annulus and F is the following function of the plasma dimensions:

$$F = (b - a)^{-2} \ln \left\{ \frac{b}{a} \left[\frac{1 + (1 + (2a/L)^2)^{\frac{1}{2}}}{1 + (1 + (2b/L)^2)^{\frac{1}{2}}} \right] \right\}. \quad (16)$$

Solving for $\bar{\rho}$ gives

$$\bar{\rho} = \overline{nM} = (1/\mu_0 F)[-B_0^3 \, \Delta B/V^2]. \quad (17)$$

The quantity in the brackets is found from the data of Fig. 10 to be $\sim 2.3 \times 10^{-9}$ in mks units over a wide range of V and B_0. The value of F for $a = 4.5$ cm, $b = 9.5$ cm and $L = 10$ cm is 1.8×10^2 m^{-2}. These values lead to a mean mass density[17] of $\bar{\rho} = 1.0 \times 10^{-8}$ g/cm^3. If the rotating ions are primarily deuterium, this corresponds to a mean ion density $\bar{n} = 3.0 \times 10^{15}$ cm^{-3}. If high-Z impurities are present in important amounts, the number density is proportionately lower. When the mass density is combined with the drift velocity of 8×10^6 cm/sec computed in the last section, one obtains a drift energy density $\frac{1}{2}\bar{\rho} v_D^2 = 0.032$ joules/cm^3. Multiplying this result by the annulus volume of 2.2×10^3 cm^3, one obtains a total drift energy of 70 joules, as compared to 60 joules deduced earlier from the magnetic probe data. The drift energy per deuterium ion is 65 ev. The previous analysis indicated that the sum of the Larmor and rotational energy was ~ 130 joules. The difference between this value and the rotational energy just deduced represents a Larmor energy of ~ 60 joules. This energy is divided between the ions and electrons in the ratio of their temperatures T_i and T_e. If we assume[18] $T_i = T_e$ and that the plasma is composed of deuterium ions, we deduce a plasma temperature of ~ 30 ev. A comparison of the foregoing quantities will be made with those obtained from a different experiment in the following paper.

V. THE OSCILLATING MODE OF OPERATION

A. Introduction

As stated previously, Ixion can be operated in two ways: (1) in the holding mode where the current remains unidirectional or (2) in the oscillating mode in which the current and voltage oscillate with time. The oscillating operation can be achieved by lowering the magnetic field or raising the Ixion voltage. The upper limit on the magnetic field that

[17] In this calculation the small contribution to ΔB arising from the induced currents in the outer electrode has been neglected. Their effect reduces the measured value of $|\Delta B|$ and therefore implies slightly higher mass densities.

[18] Evidence that the electrons have been heated is given in paper II.

will sustain the oscillatory mode is raised when external inductance is added to the circuit.

The oscillating mode of operation is a demonstration that Ixion displays the properties of a capacitor. The capacitance properties of a rotating plasma arise from the fact that a plasma in orthogonal **E** and **B** fields can display a high dielectric constant. For the uniform-field case the permittivity is given by[19]

$$\epsilon = \epsilon_0 + \rho/B^2, \qquad (18)$$

where ϵ_0 is the permittivity of vacuum. The second term represents the contribution arising from plasma drift energy, and for plasma densities such as are present in Ixion $\rho/B^2 \gg \epsilon_0$. For this reason the effective capacitance of Ixion is increased from its vacuum value of $\sim$60 $\mu\mu$f to tens of microfarads.

An exact calculation of the Ixion capacitance requires, among other things, a knowledge of the space distribution of the plasma. Formula (18) is modified in cylindrical geometry as a result of centrifugal forces of rotation. Contributions arise from the diamagnetic effect[20] which can make the capacitance nonlinear. For the present discussion, use will be made of an approximate capacitance formula derived from Eq. (18) in which ρ is replaced by an average value $\bar{\rho}$ and B is approximated by the applied vacuum field B_0. This approximate formula for the Ixion capacitance C_I is obtained by substituting ϵ from Eq. (18) (ignoring the negligible first term) into the formula for a coaxial capacitor:

$$C_I = 2\pi L\epsilon/\ln(b/a) = 2\pi L\bar{\rho}/B_0^2 \ln(b/a). \qquad (19)$$

An equivalent circuit[21] for the Ixion experiment then consists of a series connection of two capacitances (Ixion and the energizing bank), an external circuit resistance, and the sum of the external and Ixion inductances. The energy losses in Ixion can be represented by an additional shunt resistance across C_I.

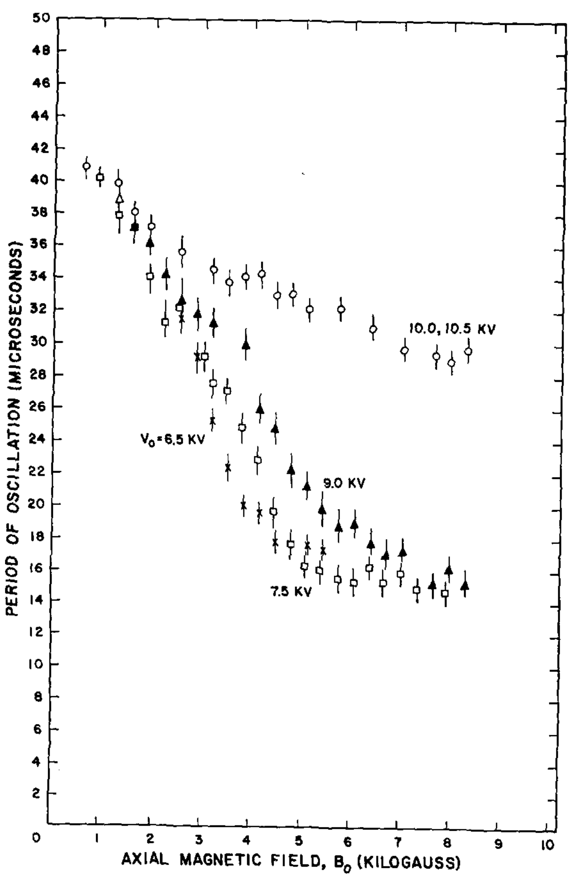


FIG. 11. Natural period of Ixion oscillation vs applied magnetic field B_0 for four values of initial voltage V_0.

B. Experimental Results

The oscillating mode was studied as follows: For a given initial applied voltage V_0, the magnetic field B_0 was varied from high values, corresponding to the holding mode, to low values at which the oscillatory mode occurred. The ringing periods were measured from the current oscillograms [see Fig. 3(c)]. The period of oscillation is plotted vs B_0 in Fig. 11. It is seen that the curves converge at $B_0 = 0$ to a common value of 42 μsec. At zero magnetic field, the capacitance of an ideal Ixion theoretically becomes infinite and its effect on the ringing period is presumed to be negligible. The effective series capacitance of the circuit is then that of the driving bank, 120 μf. One then computes the inductance of the circuit to be 0.37 μh, which agrees favorably with the value of 0.4 μh based upon external circuit measurements and the estimated Ixion inductance.

Using the equivalent circuit mentioned above, an equivalent capacitance of Ixion can be computed from the period of oscillation. The results of such a calculation, with the effects of resistance upon the period assumed negligible are given in Fig. 12, which is a plot of Ixion capacitance vs B_0 with the applied initial voltage V_0 as a parameter. The three lines drawn among the points for V_0 = 6.1, 7.5, and 9.0 kv have a slope of -2 as predicted by Eq. (19)

[19] See for example, L. Spitzer, Jr., *Physics of Fully Ionized Gases* (Interscience Publishers, Inc., New York, 1956), p. 56.

[20] A calculation of the capacitance of a rotating plasma under restrictions such as flux conservation, uniform particle density, and complete diamagnetism at the center rod, is given in reference 7.

[21] This equivalent circuit is similar to that used for the homopolar device; [O. A. Anderson, W. R. Baker, A. Bratenahl, H. P. Furth, and W. B. Kunkel, J. Appl. Phys. **30**, 188 (1959)].

and represent the trend of the data reasonably well for low values of B_0. The experimental points depart from these lines at the higher values of magnetic field where a transition to the holding mode is observed to set in. This departure may be an expression of the fact that as B_0 is increased the Ixion capacitance becomes smaller and the resistive losses are such that critical damping is approached, having the effect of lengthening the period. The Ixion capacity calculated without damping would then be higher than actual values at high magnetic fields, thus explaining the trend shown in Fig. 12. The appearance of the Ixion voltage and current curves also suggest that the difference between the oscillating and holding modes has to do with the device being under and overdamped respectively. In the case of the 10 kv data, the machine oscillates violently for all values of B_0 and there is a departure of the data from the $1/B^2$ trend obtained with lower voltages. This is undoubtedly associated with conduction across the end insulators of Ixion at these high voltages.

A calculation of the average plasma mass density from Eq. (19), using the capacitance curves as drawn in Fig. 12 and the plasma dimensions of the last section, yields $\bar{\rho} = 1.0 \times 10^{-8}$, 1.1×10^{-8}, and 1.5×10^{-8} g/cm^3 for $V_0 = 6.5$, 7.5, and 9.0 kv, respectively. This result is in remarkably good agreement with the value of $\bar{\rho} = 1.0 \times 10^{-8}$ g/cm^3, deduced above from the probe data for the holding mode. The apparent increase in density with initial bank voltage may reflect an increased number of particles which are driven from the wall at the higher electrode current densities.

One concludes from these observations that Ixion rotates in the oscillatory mode in a manner first observed by the Berkeley group[22] in their first homopolar device. This conclusion is further borne out by the fact that the diamagnetic signals observed by a probe on the axis are in the form of unidirectional pulses following the oscillating voltage.

VI. CONCLUSION

The conclusions reached from the foregoing experiments may be summarized as follows: For the range of magnetic fields studied, the Ixion charging current pulse imparts an energy to the plasma of ~130 joules. This energy is divided nearly equally between rotational drift and thermal motion. The Ixion voltage drops to a value such that the drift velocity has a value of $\sim 8 \times 10^6$ cm/sec, nearly independent of the value of applied magnetic field. If this energetic plasma is assumed to be composed primarily of deuterium and electrons with the same temperature, the ion drift energy and plasma temperature are ~65 ev and ~30 ev, respectively. The rotating plasma, after being highly compressed toward the midplane during the current pulse, relaxes into an annular volume of approximately two liters extending radially ~5 cm from the effective center electrode and axially ~5 cm each side of the Ixion midplane. At an average magnetic field of 8 kgauss the β value of the plasma is about 12%. Computations of the mean plasma density from both the plasma diamagnetism and the natural period of Ixion when operated as an oscillating hydromagnetic capacitor yield the value $\bar{\rho} \approx 1 \times 10^{-8}$ g/cm^3. The corresponding number density for a deuterium plasma is 3×10^{15} ions/cm^3.

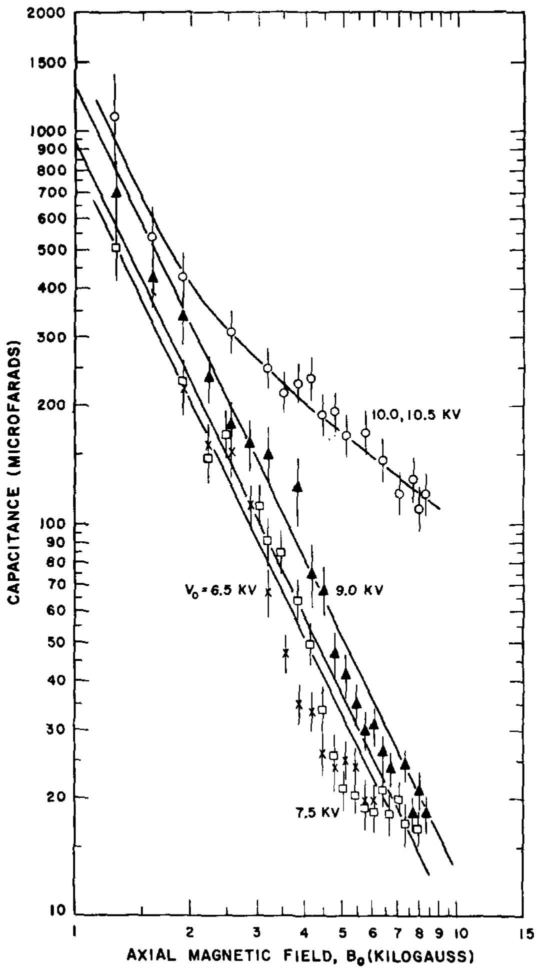


FIG. 12. Ixion capacitance as deduced from the period of oscillation.

[22] O. A. Anderson, W. R. Baker, A. Bratenahl, H. P. Furth, J. Ise, Jr., W. B. Kunkel, and J. M. Stone, in *Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy* (United Nations, Geneva, 1958), Vol. 32, p. 155.

The energy supplied from the charging bank during the establishment of the above conditions decreases with increasing values of applied magnetic field. After rotation is established the voltage is drained from the capacitor bank at a rate which, for a given magnetic field, increases with the value of the Ixion voltage. Ixion is indeed lossy from an energy point of view. Experimental investigations of the energy losses have been made and are discussed in the following paper.

The analysis of the experimental results required many assumptions, none of which are exempt from some doubt. For example, the assumption of azimuthal symmetry may only be valid in a time averaged sense. The Berkeley group[23] have found that the radial current density in the homopolar device is not uniform but flows in localized regions suggestive of rotating spokes. Confinement was also assumed rather than experimentally proven. One might at first sight conclude from the fact that active plasma is present for hundreds of microseconds under rotating conditions, as compared with 20 μsec when the electric field is removed, that an enhanced confinement is evident. It must be borne in mind, however, that when an electric field is present there is the possibility of a continued ionization of neutral gas flowing in from the wall and end regions. The plasma may be kept active by constantly replenishing the particles lost with newly ionized ones. A plasma device that can rejuvenate itself in such a fashion may keep active plasma present for a long time by drawing its energy from the electric field sources without necessarily having a high degree of confinement.

[23] A. Bratenahl and W. B. Kunkel, Controlled Thermonuclear Research Quart. Rept., UCRL-8775 (1959), p. 54. Also A. Bratenahl and K. Halback, Controlled Thermonuclear Research Quart. Rept., UCRL-8887 (1959), p. 79.

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